⌈Riesz representation Theorem.

⌈Thm. Let V be a fin. dim. inn. prod. v.s.
For any linear functional $f: V \to F$, there is a unique vector $\beta \in V$ s.t.

$$\text{for all } \alpha \in V, \quad f(\alpha) = \langle \alpha, \beta \rangle$$

In other words, $\varphi: V \to V^*: \beta \mapsto \langle \cdot, \beta \rangle$ is a Conjugate isomorphism.
Proof. Let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal basis for V.
Given linear functional $f: V \to F$, put

$$\beta = \sum_{j=1}^{n} \overline{f(\alpha_j)} \cdot \alpha_j.$$

then, for each $1 \le k \le n$,

$$\langle \alpha_k, \beta \rangle = \left\langle \alpha_k, \sum_{j=1}^{n} \overline{f(\alpha_j)}\, \alpha_j \right\rangle = \sum_{j=1}^{n} f(\alpha_j) \cdot \langle \alpha_k, \alpha_j \rangle = f(\alpha_k)$$

Clearly, $\langle \cdot, \beta \rangle$ is linear, thus the existence is proved. And, the uniqueness is shown:
if $\langle \alpha, \beta \rangle = \langle \alpha, \gamma \rangle$ for all $\alpha \in V$, then for $\alpha = \beta - \gamma$, $\langle \beta - \gamma, \beta - \gamma \rangle = 0$.

Thus, $\beta = \gamma$. ☒

Observe that a inn. prod space V with orthonormal basis $\{\alpha_1, \cdots, \alpha_n\}$.
Let $\alpha = x_1\alpha_1 + \cdots + x_n\alpha_n$ and $\beta = y_1\alpha_1 + \cdots + y_n\alpha_n$. Then,

$$\langle \alpha, \beta \rangle = x_1\overline{y_1} + \cdots + x_n\overline{y_n}$$

Meanwhile, for any linear functional $f \in V^*$,

$$f(\alpha) = f(\alpha_1)x_1 + \cdots + f(\alpha_n)x_n$$

If we want to β satisfies $\langle \alpha, \beta \rangle = f(\alpha)$, then take $y_i = \overline{f(\alpha_i)}$. That is,

$$\beta = \overline{f(\alpha_1)}\,\alpha_1 + \cdots + \overline{f(\alpha_n)} \cdot \alpha_n$$

「Geometric Observation.

Given a linear functional $f: V \to F$, $V = \ker f \oplus (\ker f)^{\perp}$.

Let P be the orthogonal projection of V on $(\ker f)^{\perp}$. Then,

$$\text{for all } \alpha \in V, \quad f(\alpha) = f(P(\alpha)).$$

Now, if $f \neq 0$, then $\operatorname{rank} f = 1$, so for any non-zero vector $\gamma \in (\ker f)^{\perp}$,

$$P(\alpha) = \frac{\langle \alpha, \gamma \rangle}{\|\gamma\|^2} \cdot \gamma \quad \text{for all } \alpha \in V.$$

Finally, for given $\alpha \in V$,

$$f(\alpha) = f(P(\alpha)) = \langle \alpha, \gamma \rangle \cdot \frac{f(\gamma)}{\|\gamma\|^2}, \quad \text{and } \beta = \frac{\overline{f(\gamma)}}{\|\gamma\|^2} \cdot \gamma.$$